

Interactive Learning in Microelectronics Education: Comparing PC and Mixed Reality Approaches for Student Engagement and Visual-Spatial Memory

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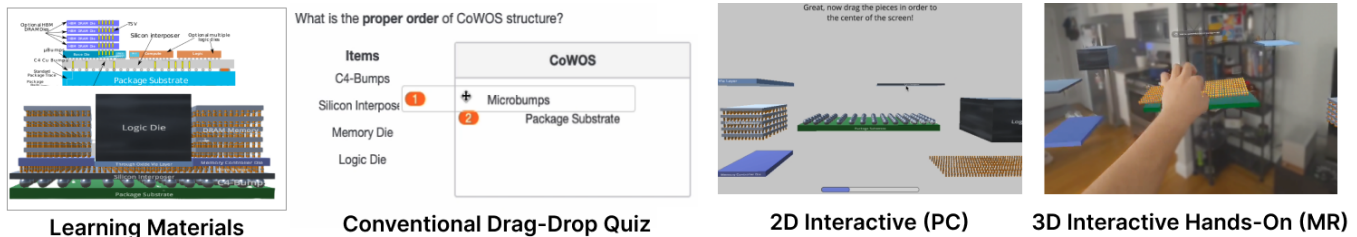


Figure 1: Comparison of learning approaches. Left: Conventional method utilizing text-based quizzes. Middle: Interactive, Lego-like learning experience conducted on a PC. Right: Hands-on interactive learning facilitated by mixed reality (MR).

Abstract

The microelectronics industry continues to face a persistent talent gap, highlighting the need for effective learning modules to engage future engineering students. Mixed Reality (MR) offers a promising hands-on approach for teaching complex visual-spatial structure of microelectronic; however, whether MR-based instruction outperforms conventional desktop interfaces in enhancing visual-spatial memory for microelectronics education remains unclear. In this preliminary study, we developed an interactive “LEGO-like” assembly experience in both MR and PC formats to compare their effects on students’ learning experiences. Visual-spatial knowledge was assessed through three quiz tasks—component name recognition, visual selection, and spatial ordering—administered before the intervention, immediately afterward, 24 hours later, and again at six weeks to evaluate retention. Preliminary findings suggest that PC users exhibited greater immediate knowledge gains (PC: 89%, MR: 73%), whereas MR users demonstrated stronger retention at the six-week mark. We discuss these results and directions for future research.

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CCS Concepts

• **Human-centered computing** → **Interaction Design**; *Empirical studies in interaction design*.

Keywords

Mixed Reality, STEM Education, Comparison, Knowledge Retention

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1 Introduction

Enhancing student engagement and improving learning outcomes in engineering education has been a longstanding focus within the Human-Computer Interaction (HCI) and Engineering Education research communities [23, 40, 61]. This challenge is particularly pronounced in microelectronics manufacturing, a field that continues to face a critical talent shortage in the United States despite historical investments and recent legislative support, such as the CHIPS and Science Act of 2022 [12, 52]. Our prior survey of U.S. engineering students and faculty in microelectronics highlights that the inherent complexity of the subject matter, combined with limited opportunities for hands-on training, presents significant barriers to

student engagement compared to other engineering disciplines [24]. For example, mastering advanced chip packaging requires intricate spatial reasoning, an understanding of complex interdependencies between layers, and precise procedural knowledge for component assembly—skills that are difficult to effectively convey through traditional two-dimensional, image-based lectures [38]. Immersive technologies, such as Mixed Reality (MR), present a promising approach to addressing these challenges by enabling interactive, hands-on learning experiences [49]. Prior research suggests that immersive technologies are particularly effective for educational tasks involving visual-spatial skills and procedural knowledge—both of which are fundamental to microelectronics manufacturing [17, 43]. Nonetheless, whether MR can produce better learning outcomes than conventional desktop-based interfaces in microelectronics education remains insufficiently understood.

Previous research on immersive technologies (e.g., VR or MR) in education generally shows they enhance student engagement and enjoyment, yet their impact on learning outcomes is less clear [1, 9, 25]. For instance, a meta-review by Chen et al. [9] indicates that while VR has been shown to significantly enhance engagement (effect size > 0.4), its effect on measurable learning outcomes, such as test performance, is less definitive. The recently proposed Cognitive Affective Model of Immersive Learning (CAMIL) by Makransky and Petersen [30] suggests that such mixed findings stem from variations in instructional methods across studies, including differences in learning contexts and the timing of learning assessments. Most prior studies primarily measure learning outcomes immediately after the intervention, dedicating less attention to long-term knowledge retention [19, 34]. This limited focus on sustained retention is especially concerning in microelectronics education, where visual-spatial memory of its structural relationship is crucial [38]. Investigating whether interactive MR can improve visual-spatial memory in microelectronics is therefore essential.

To address this gap, we developed an interactive learning experience in which students assemble chip packaging structures using a LEGO-like method, commonly employed to teach visual-spatial relationships and assembly procedures [16]. Two versions of this learning tool were created: (1) a PC-based interface operated with a mouse, and (2) a Mixed Reality (MR) interface enabling direct manipulation of virtual components by hand. This study examines three key research questions:

- **RQ1:** Does hands-on interaction in MR lead to greater gains in visual-spatial knowledge compared to PC-based interaction?
- **RQ2:** Does hands-on interaction in MR result in better long-term retention of visual-spatial knowledge than PC-based interaction?
- **RQ3:** Do engineering students find MR more useful and indicate a higher intention to adopt it than PC?

This study contributes to the fields of Human-Computer Interaction (HCI) and Engineering Education in several key ways. First, while immersive technologies such as Mixed Reality (MR) have been widely studied in STEM education, much of the existing research has focused on disciplines such as organic chemistry, mathematics, and medical training [14, 34]. By contrast, the application of immersive technologies in microelectronics education

remains underexplored. To address this gap, the MR applications developed in this study have been made publicly accessible via the Meta Store and GitHub, facilitating broader adoption and enabling future research in this domain. Second, comparative studies evaluating different learning interfaces (e.g., PC vs. MR) in HCI research often assess learning outcomes immediately after the intervention, with limited emphasis on long-term knowledge retention [19, 26]. This study aims to address this gap by measuring retention at two post-intervention time points: 24 hours and six weeks. Preliminary findings suggest a trend where PC-based learning may be more effective in enhancing immediate learning outcomes, whereas MR appears to be more effective in maintaining knowledge retention over a longer period. Finally, this research builds upon embodied cognition theory by exploring the relationship between sense of agency in MR and memory retention, particularly investigating its potential moderating role in knowledge outcomes. Although the small sample size did not yield statistically significant findings, the overall results lay the groundwork for future research on how MR can be strategically utilized to enhance educational outcomes in microelectronics.

2 Related Works

2.1 Mixed Reality in Microelectronics Education

Microelectronics is a branch of electronics focused on the design, fabrication, and manufacturing of extremely small electronic components and circuits [64]. Microelectronic devices play a crucial role in advancing modern technologies, including artificial intelligence, quantum computing, and the Internet of Things, as their packaging structure and design are directly linked to improved efficiency in electronic systems [48]. Despite its significance, faculty in this field have reported difficulty engaging engineering students, partly due to the inherent complexity of the subject, which requires an understanding of materials science, physics, chemistry, and electrical engineering [24]. As immersive technologies such as VR, AR, and MR gain traction for enhancing hands-on learning and student engagement in engineering education [55], researchers are beginning to explore their use in microelectronics [49]. For instance, Cornell's NanoScale Science Facility offers VR-based cleanroom tours [58], and Purdue University has partnered with Intel to develop a virtual fabrication environment called VFabLab for safe equipment operation training [44, 60]. Additionally, Lee [28] created a VR cleanroom training program for vocational education, reporting positive feedback from field experts. However, these efforts have largely focused on procedural task training (e.g., machine operation) rather than conceptual knowledge, such as visualizing the spatial arrangement of microelectronic components.

Visual-spatial reasoning—the capacity to interpret and analyze component relationships through visual and spatial cues [32]—is essential skills in learning microelectronics [37]. Numerous studies on immersive technologies have shown that the visual-spatial affordances and interactivity of VR/MR significantly enhance spatial reasoning skills [62]. For example, Cai, Chiang, and Wang [8] introduced AR techniques in a physics course, where students interacted with 3D convex imaging models, significantly improving their spatial reasoning in determining light travel paths in convex imaging experiments. Behmke et al. [7] examined whether mobile-based AR

helped students grasp stereoisomers—molecules that share a molecular formula but differ in 3D arrangement—in organic chemistry. Altmeyer et al. [2] looked specifically at how AR could support conceptual knowledge construction, designing a tablet-based AR system that overlaid relevant information on a physical circuit board. Their findings indicated significant conceptual gains only for the AR-supported approach, suggesting that immersive technologies like MR can also enhance conceptual learning outcomes and not just procedural tasks.

2.2 Prior Work Comparing Immersive Interfaces and PC-Based Learning

Comparing the impact of different media on learning experiences has been a popular research topic in both the HCI and educational technology [25, 29, 31]. However, studies have reported mixed outcomes on the efficacy of immersive interfaces (e.g., VR, AR, MR) relative to PC-based methods in measurable learning gains. For example, Makransky et al. [29] conducted a study with 165 high school students, comparing the effects of VR and conventional video on two types of knowledge: declarative (e.g., multiple-choice questions assessing conceptual or factual information) and procedural (e.g., open-ended questions requiring a sequence of steps). Drawing on motivational theories of learning [45, 51], which posit that students learn more effectively when they are engaged and enjoy the process, they hypothesized that VR would increase enjoyment and presence, leading to improved outcomes for both knowledge types. Contrary to their expectations, while VR did enhance enjoyment and presence, it did not produce better learning outcomes compared to video-based instruction. In another study, Makransky et al. [31] compared an interactive, immersive science lab (delivered via Samsung Gear VR) to a PC-based version of the same content. Their results indicated that students learned significantly less in the VR condition. Additional analysis of students' EEG data suggested that the immersive VR environment introduced extraneous cognitive load, making it more difficult for students to concentrate on the learning material than when using a traditional desktop interface.

These findings differ from numerous other studies suggesting that immersive interfaces (AR, VR, and MR) can yield better learning outcomes than PC-based instruction [19, 34]. Although variations in study design make it challenging to draw definitive conclusions about the impact of immersive technology on learning outcomes, these results indicate that effectiveness is not determined solely by the medium (VR, MR, or PC). Rather, instructional design—what content is taught and how outcomes are assessed—also plays a significant role [30]. Moreover, a recent systematic review highlighted a key limitation in prior research comparing PC-based and immersive learning technologies: most studies measured learning outcomes only immediately after the intervention, with relatively few exploring long-term knowledge retention. Given the importance of sustained retention for educational effectiveness, further research is needed to examine the long-term impact of immersive technologies on learning outcomes [19].

2.3 Mixed Reality and Knowledge Retention

Knowledge retention, the ability to store and recall learned information over time, is widely considered a crucial factor in education [4].

Ebbinghaus's early work demonstrated that individuals forget over half of newly acquired information within an hour, emphasizing the importance of strategies to improve retention [15]. Multiple theoretical foundations, including embodied cognition [53] and dual coding theories [41, 42], provide valuable groundwork for understanding how hands-on experiential learning through MR can enhance knowledge retention. According to Paivio's Dual Coding Theory, the human brain processes information through two interconnected systems: one for verbal inputs (words and text) and another for nonverbal inputs (images and spatial representations) [41, 42]. When abstract concepts are presented both visually and verbally, information processing and storage are facilitated [33]. Numerous studies support this idea, showing that multimedia formats integrating visual-spatial cues with textual cues can enhance learning outcomes [57]. Because MR allows for interactive 3D visuals coupled with verbal information, it has the potential to support dual coding and increase memory retention. In addition, research grounded in embodied cognition theory suggests that cognitive processes are closely tied to physical interactions with the environment, highlighting the importance of hands-on activities for long-term memory [27, 54, 59]. Within immersive technology, the sense of agency—awareness of and control over one's actions and outcomes—is known to be linked to cognition and memory [18, 36]. By actively manipulating objects or navigating virtual scenarios with a high degree of control, learners engage in embodied cognition, using sensorimotor experiences to construct and retrieve knowledge [50, 63].

Despite these theoretical foundations, findings on the impact of immersive technology on knowledge retention remain inconclusive. For instance, Babu et al. [5] found that the VR group had better knowledge retention compared to the tablet computer interface when the test was re-administered one day later, although participants performed similarly on a mechanical labeling task using either VR or a tablet computer immediately after the intervention. In contrast, Harrington et al. [20] conducted a study with medical students and found no significant differences in knowledge retention scores between those who viewed information through an HMD or a traditional television screen. Given that learning outcomes can vary depending on the specific content and field of study, these inconsistent findings highlight the need for further research.

3 Design and Methodology

3.1 Learning Contents and Interaction Design

The specific learning topic focuses on the structure of CoWoS (Chip-on-Wafer-on-Substrate) packaging, a cutting-edge technology currently utilized by TSMC [22]. The structure comprises various elements, including the substrate, C4 bumps, interposer, microbumps, memory die, DRAM memory, logic die, and more. We chose this structure because CoWoS represents a significant advancement in semiconductor manufacturing, enabling the integration of multiple chips on a single wafer with high-density interconnect substrates, thereby making it a critical topic for microelectronics education. The learning content was carefully designed by two Ph.D. students, whose expertise lies in the topic, along with instructors who currently teach the introductory course on semiconductor packaging. In alignment with the course learning objectives, the content aims

to teach the name of each component and the visual-spatial structure of these components by prompting students to memorize the correct name of each component, its spatial relationships (e.g., order), and its proper location within the structure. Building upon previous work that utilized LEGO-assembly models for spatial cognitive education tasks [21], the learning experience is divided into two parts. First, users can learn about each element by toggling its visibility and zooming in and out to explore details. Specifically, users are prompted to click all toggle buttons to learn the name of each component and its spatial structures. Users can closely examine a component's structure by scaling the 3D chip object. Second, users are prompted to complete an assembly task in which they need to rearrange the randomly scattered elements into the correct structure. When users place a component in the wrong spot, an error message appears with a sound indicating that the answer was incorrect. Participants can repeat the process as many times as needed until they successfully complete the assembly in the correct structure.

With respect to interaction design, in the PC version, users interact with the interface using a mouse to toggle elements and drag-and-drop components into their appropriate positions. In the MR version, users interact using their hands; they can grab and assemble components and press the toggle button using hand gestures. Following Mayer's Cognitive Theory of Multimedia Learning and instructional design principles [33], the interface integrates audio and textual information alongside 3D visual elements to enhance the overall learning experience.

3.2 Apparatus

The interactive learning content was developed using Unity3D (version 2022.3.40f). The 3D model of the CoWoS structure was created and animated in Maya, and interactions were scripted in C#. The PC version was published as a WebGL build with a resolution of 1024 x 768. The MR experience was developed using the Unity XR Interaction Toolkit (v3.0.7), with hand interaction functionality implemented via XR Hands (v1.4.3). The final deliverable is compatible with multiple consumer-grade VR/MR headsets, including the Quest Pro, Quest 3, and Quest 2. For this evaluation study, we utilized the Meta Quest 3.

3.3 Participants

For this preliminary study, 27 engineering students were recruited through convenience sampling. We specifically sought students with no prior knowledge of the subject matter, as existing knowledge could influence learning outcomes [35]. One participant did not complete the survey, resulting in a final dataset of 26 participants. The study employed a between-subject experimental design, with 12 participants assigned to the MR group and 14 to the PC group. We aimed to ensure a balanced distribution across gender, education level, and majors in both groups. The majority of participants were between the ages of 25–34 (40%), followed by 18–24 years old (33%). In terms of gender distribution, 55% of participants were male, and 45% were female. Regarding education level, 46% were undergraduate students, and 53% were graduate students. Most participants were Computer Science majors (65%), followed by Chemical Engineering (20%), Mechanical Engineering (20%),

Electrical Engineering (13%), and other engineering fields, such as Biomedical Engineering (6%). Given the exploratory nature of this study, we did not specifically control for participants' prior experience with MR and VR to obtain a more representative sample of engineering students who might register for the introductory semiconductor packaging course. 58% of participants reported prior experience using HMDs. The study was reviewed and approved by the University of Florida's Institutional Review Board (IRB).

3.4 Procedure

Participants completed the study in four phases. In the first phase, all participants completed a pre-survey to assess their baseline knowledge of CoWoS packaging and to collect demographic information. This was followed by the learning module, where participants engaged with the educational content in their assigned environments. Participants in the MR condition used a Quest 3 headset to explore and learn about CoWoS, while the PC group accessed the same content on a desktop computer. After completing the learning module, participants proceeded to the second phase by completing a post-survey. This survey assessed their immediate visual-spatial memory through visual-spatial cognition tasks via quizzes, requiring them to identify the correct CoWoS structure and recognize component names based on what they had learned in the interactive modules. The survey also evaluated their experiences, including perceived usability, usefulness, engagement, and other factors, as detailed in the subsection below. In the third phase, 24 hours after the initial learning session, participants were sent a follow-up survey with the same quiz questions. In the fourth phase, participants completed the quiz again, identifying the name of each component and selecting the correct CoWoS structure to assess knowledge retention.

3.5 Measures

The primary dependent variable is *learning outcome*, which was measured through a quiz. In addition to learning outcomes, other dependent measures include *engagement*, *sense of agency*, *usefulness*, and *future intention to use*. Furthermore, *perceived usability* was also measured to account for its impact as a confounding variable. *Total time spent* on the learning task was included as a control variable. Learning outcomes were measured at various time interval to assess the retention, while other measures are measured once immediately after the intervention.

Learning Outcomes Learning outcomes were assessed through a quiz that consisted of three tasks. The first task was a *visual selection* exercise, where participants selected the correct image of CoWoS from five options. The second was a *spatial task*, requiring participants to reorganize randomly distributed CoWoS elements into the correct order. The final task was a *recognition task*, in which participants identified the correct names of CoWoS elements from multiple-choice options. These outcomes were measured at four different time points: *before learning*, to establish a baseline of initial knowledge; *immediately after learning*, to measure immediate knowledge gain; *24 hours later*, to evaluate short-term retention; and *6 weeks later*, to assess long-term retention. Knowledge gain (g_i) was calculated following Colt et al.'s [11] normalized approach to account for individual differences in prior knowledge.

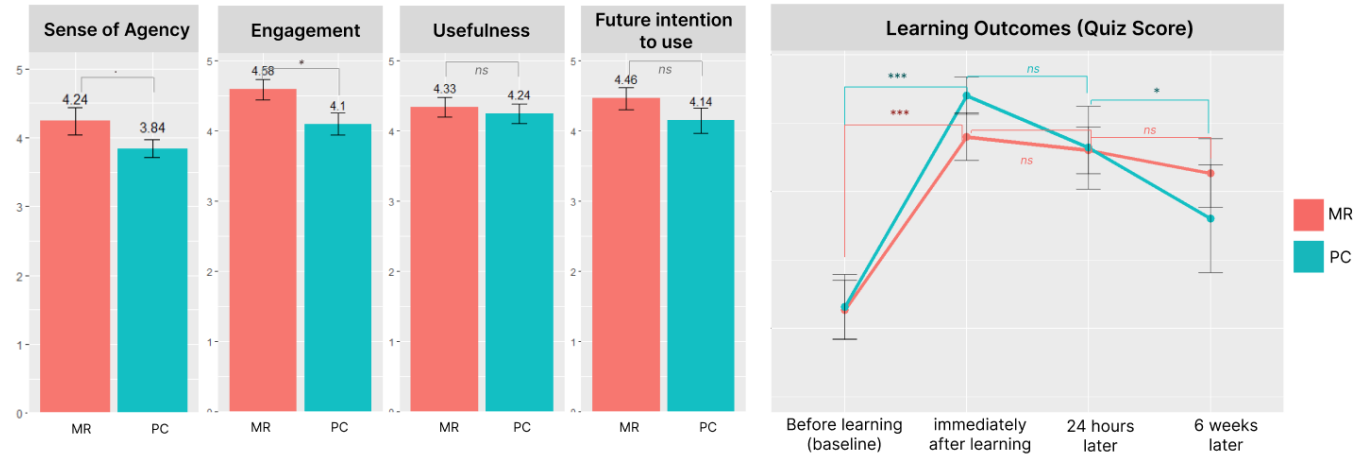


Figure 2: Comparison of MR and PC on engagement, perceived usefulness, future intention to use, and learning outcomes (quiz scores). Quiz scores were measured at four time points: baseline (before learning), immediately after learning, 24 hours later, and 6 weeks later (note: ** $p < .001$, *** $p < .01$, ** $p < .05$, 'ns' not significant).**

The pre-intervention quiz score served as the baseline for determining knowledge gain across different intervals.

$$g_i = \frac{\text{post-test score} - \text{baseline score}}{100 - \text{baseline score}}$$

Engagement was assessed using three questionnaire items adapted from O'Brien et al.'s *User Engagement Scale* [39], such as "This learning experience was fun," "I was deeply involved in this learning experience," and "This learning experience was engaging." The engagement scale demonstrated good internal consistency, with a *Cronbach's alpha* value of 0.798 [56].

Sense of Agency The sense of agency was measured using four questionnaire items adapted from Pritchard [47] and Polito's *Sense of Agency Rating Scale (SOARS)* [46]. Participants rated statements such as "I felt a strong sense of control over the learning experience," "I felt that my actions and decisions were under my control," and "I was in control of the target object." The *Cronbach's alpha* value for this scale was 0.844, indicating strong internal consistency.

Usefulness and Future Intention to Use Perceived usefulness and future intention to use were measured based on the Technology Acceptance Model (TAM) [13]. Usefulness was assessed using three questionnaire items: "This learning tool will be useful for students," "This learning tool would enhance learning outcomes for students," and "Using the tool enhanced my understanding of the subject matter." Future intention to use was measured with two items: "I would like to use this tool for future learning" and "I would recommend this tool to other learners." *Cronbach's alpha* for both measures exceeded .70 [56].

Usability (Control Variable) The usability of the interface was included as a control variable, as the perceived difficulty of using the MR interface could affect students' engagement and attention levels. Usability was measured using a modified version of the *System Usability Scale (SUS)* [6], with items such as "I thought this app was easy to use" and "I found this app unnecessarily complex."

Total Time Spent (Control Variable) Total time spent on the learning task was controlled for, as it is known to be significantly correlated with student engagement and learning outcomes. Participants in the PC condition were required to spend a minimum of 3 minutes and 30 seconds on the interactive learning tasks, a duration determined based on the average time spent by participants in the MR condition. The total time spent on the learning interaction did not differ significantly between the PC group ($M = 265.44$ s) and the MR group ($M = 226.70$ s; $t = -0.943$, $df = 19.76$, $p = 0.357$), supporting the equivalence of engagement time across conditions.

4 Results

4.0.1 Control Variables. To control for potential confounding effects, usability scores were included as a covariate in subsequent analyses. Another important confounding variable is students' prior knowledge. However, a t-test analysis of baseline quiz scores revealed no statistically significant differences in baseline task performance between the PC and MR groups, suggesting that both groups had comparable levels of prior knowledge ($p = .533$).

4.0.2 Learning and Retention. Figure 2 presents the study findings on user evaluations and changes in quiz performance. To examine changes in learning outcomes between the MR and PC conditions, knowledge gain was compared across three assessment points: immediately after learning, 24 hours later, and six weeks later, relative to the baseline. A series of pairwise comparisons were conducted to evaluate knowledge gain at each time point. For *immediate learning*, a statistically significant increase in quiz scores was observed for both the PC condition ($t(10) = -6.70$, $p < .0001$, $d = -2.62$) and the MR condition ($t(11) = -5.15$, $p < .0001$, $d = -1.47$), indicating significant knowledge gains in both conditions compared to the baseline. To further compare knowledge gain between MR and PC, an ANCOVA was conducted, with *condition (MR vs. PC)* as the independent variable, *usability as a covariate*, and *knowledge gain as the dependent variable*. Although PC users demonstrated higher immediate knowledge gain ($M = 89.1\%$, $SD = 19.2$) compared to MR

users ($M = 73.8\%$, $SD = 24.5$), the difference was only marginally significant ($F(1, 24) = 3.198$, $p = .086$), with a *moderate effect size* ($d = -0.704$) based on Cohen's guidelines[10].

For *knowledge retention at 24 hours*, there was no statistically significant difference between MR and PC conditions ($F(1, 21) = 0.159$, $p = .694$), with a small effect size ($d = 0.167$). Similarly, at the *six-week assessment*, no statistically significant difference was found between MR and PC conditions ($F(1, 20) = 1.051$, $p = .318$), though the effect size was larger ($d = 0.44$), suggesting a small-to-moderate effect. To further examine *knowledge retention over time*, a series of pairwise comparisons were conducted to compare quiz performance at different time intervals: (1) *24 hours vs. immediately after learning*, and (2) *six weeks vs. 24 hours post-learning*. While there were no statistically significant differences in quiz scores between *immediately after learning and the 24-hour follow-up* in either condition, a significant decrease in quiz scores was observed in the *PC condition at the six-week assessment* ($t(10) = 3.67$, $p < .05$, $d = 0.36$), suggesting a decline in knowledge retention over time. In contrast, no statistically significant difference was found between quiz scores at the *six-week and immediate post-learning assessments in the MR condition* ($t(11) = 2.11$, $p > .05$, $d = 0.29$), indicating *relatively stable knowledge retention* among participants who learned with MR.

4.0.3 Sense of Agency, Engagement, Usefulness, Intention to Use. A series of ANCOVA analyses were conducted to compare PC and MR, with usability included as a covariate. The results for sense of agency showed that the mean score for the MR condition ($M = 4.24$, $SD = 0.68$) was higher than that for the PC condition ($M = 3.85$, $SD = 0.47$). However, the statistical analysis revealed only a marginal difference between the two groups, $F(1, 25) = 3.065$, $p = .0928$, with a small effect size ($d = 0.11$). In contrast, for engagement, the results revealed a statistically significant difference between conditions, $F(1, 25) = 4.811$, $p < .05$, with the MR condition showing significantly higher engagement ($M = 4.58$, $SD = 0.51$) compared to the PC condition ($M = 4.10$, $SD = 0.60$), with a small effect size ($d = 0.17$). When it comes to usefulness and intention to use, the ANCOVA results indicated no statistically significant differences for either variable. For perceived usefulness, no significant differences were found between the MR and PC conditions, $F(1, 24) = 0.240$, $p = .629$, with a negligible effect size ($d = 0.09$). Similarly, for future intention to use, no significant difference was observed between the MR and PC conditions, $F(1, 25) = 1.726$, $p = .201$, with a small effect size ($d = 0.067$). These findings suggest that perceived usefulness and future intention to use the learning modules were comparable across both conditions.

4.0.4 The Impact of Sense of Agency on Knowledge Retention.

A moderated regression analysis was conducted to examine the moderating role of sense of agency in the relationship between media condition (MR vs. PC) and knowledge retention. Knowledge retention was measured as the ratio of correct responses in the follow-up quiz relative to the previous assessment. For the 24-hour follow-up, the overall model was statistically significant, $F(3, 19) = 3.65$, $p = .031$, accounting for 36.51% of the variance in knowledge retention ($R^2 = .365$). However, the interaction between media condition (PC vs. MR) and sense of agency was only marginally significant ($B = 0.65$, $t(19) = 2.09$, $p = .059$). For the six-week

follow-up, both the overall model and the interaction between media condition and sense of agency were statistically significant, indicating that sense of agency did not play a significant moderating role in enhancing knowledge retention for MR compared to PC at the six-week mark.

5 Discussion

This study explored the potential of Mixed Reality (MR) to enhance student engagement and support long-term learning outcomes in microelectronics manufacturing education, a field that remains complex and challenging due to its highly technical nature. While the small sample size in this exploratory study limits the ability to draw definitive conclusions, the preliminary findings suggest that students who engaged with the PC interface tended to achieve greater immediate knowledge gains compared to those who used MR with hands-on interaction. However, MR appeared to elicit significantly higher engagement than the PC interface. These observations are consistent with prior research by Makransky et al. [29, 31], which suggests that immersive environments such as VR can introduce extraneous cognitive load due to multiple simultaneous stimuli, potentially making it more difficult for learners to focus on instructional content compared to PC-based methods.

At the same time, this study expands on previous work by indicating that hands-on interaction in MR may contribute to long-term knowledge retention, aligning with theories of embodied cognition and experiential learning [27, 53]. While the analysis of sense of agency as a potential factor in knowledge retention did not yield statistically significant results, this may be due in part to the small sample size. Additionally, it is possible that other cognitive or instructional mechanisms, beyond sense of agency, influence knowledge retention in MR learning environments. Future research should further examine these factors to clarify the conditions under which MR supports sustained learning.

5.1 Limitation and Future Research

Several limitations should be considered when interpreting these findings. First, given the exploratory nature of this study, the sample size was small and recruited through convenience sampling. This approach may introduce selection bias and limit the generalizability of the results. Additionally, the small sample size reduces statistical power, which could obscure potential effects that might be detected in a larger study. Second, the study employed a between-subjects design without full random assignment, which may have introduced confounding variables. Individual differences among students, including motivation, interest in the subject, prior exposure to interactive technologies, or familiarity with MR, could have influenced learning outcomes [30]. Future studies should control for these variables to allow for a more rigorous comparison between MR and PC-based learning. Third, the study was conducted in a controlled lab environment rather than an authentic classroom setting. As a result, the findings may not fully capture the nuances of real-world classroom dynamics, instructional constraints, or the challenges associated with integrating MR in large-scale settings.

From a technical standpoint, the MR learning tool developed in this study relies on hand-based interaction, which may pose accessibility challenges for individuals with physical disabilities, such as

hand tremors or limited dexterity. Future iterations of MR learning tools could explore alternative interaction methods, such as voice commands or adaptive input mechanisms, to ensure greater inclusivity. Lastly, this study focused specifically on microelectronics education, and the extent to which these findings generalize to other engineering disciplines remains uncertain. However, given that MR-based learning has the potential to enhance visual-spatial reasoning, its applications may extend beyond microelectronics to other fields that require spatial understanding, such as organic chemistry, civil engineering, geophysics, trigonometry, and mechanical engineering.

Building on these preliminary findings, future research can take several directions to further explore the potential of MR in engineering education. First, expanding the study with a larger and more diverse participant pool, as well as employing randomized controlled trials, would enhance the reliability and generalizability of the results. Second, conducting studies in real classroom environments across multiple course modules would provide valuable insights into the practical integration of MR technology within engineering curricula. Additionally, further research should also examine strategies for minimizing cognitive load in immersive learning environments to optimize their effectiveness. Incorporating instructional scaffolding, adaptive feedback mechanisms, and personalized learning pathways may help students focus on essential content while reducing potential distractions [3]. By addressing these areas, future work could provide more comprehensive insights into how MR can be effectively utilized to improve engagement and support long-term learning in engineering education.

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